Global Industrial Architecture
Selected Case Study: National Archives Preservation Facility

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Group - Maisie, Noor, Sam, Ornela

Site Analysis Stage 1

Site Analysis

Typology - Ecological Function: Ecological function of vegetation on the site, such as providing habitat for wildlife, improving soil stability, enhancing biodiversity, and mitigating environmental impacts such as erosion and stormwater runoff for the facility.

Address: 30 Vicars St, Mitchell ACT 2911

Pedestrian circulation through car park.

Pedestrian /worker circulation around facility.

Sun Path and Climate:The ACT typically sees daily high temperatures in summer ranging from 26-28°C in the northern regions to approximately 18-20°C in the alpine areas. Winter brings average low temperatures ranging from 2-4°C in the north to around -4 to -2°C in the southern parts and the alpine regions. Throughout the year, this territory vast and distinct seasonal changes.

Analysis of Parking Facilities: The parking options are available next to the facility, including on-site parking areas and nearby public parking lots and garages. There is a large capacity of space, the layout enables more e=than enough access into the building.

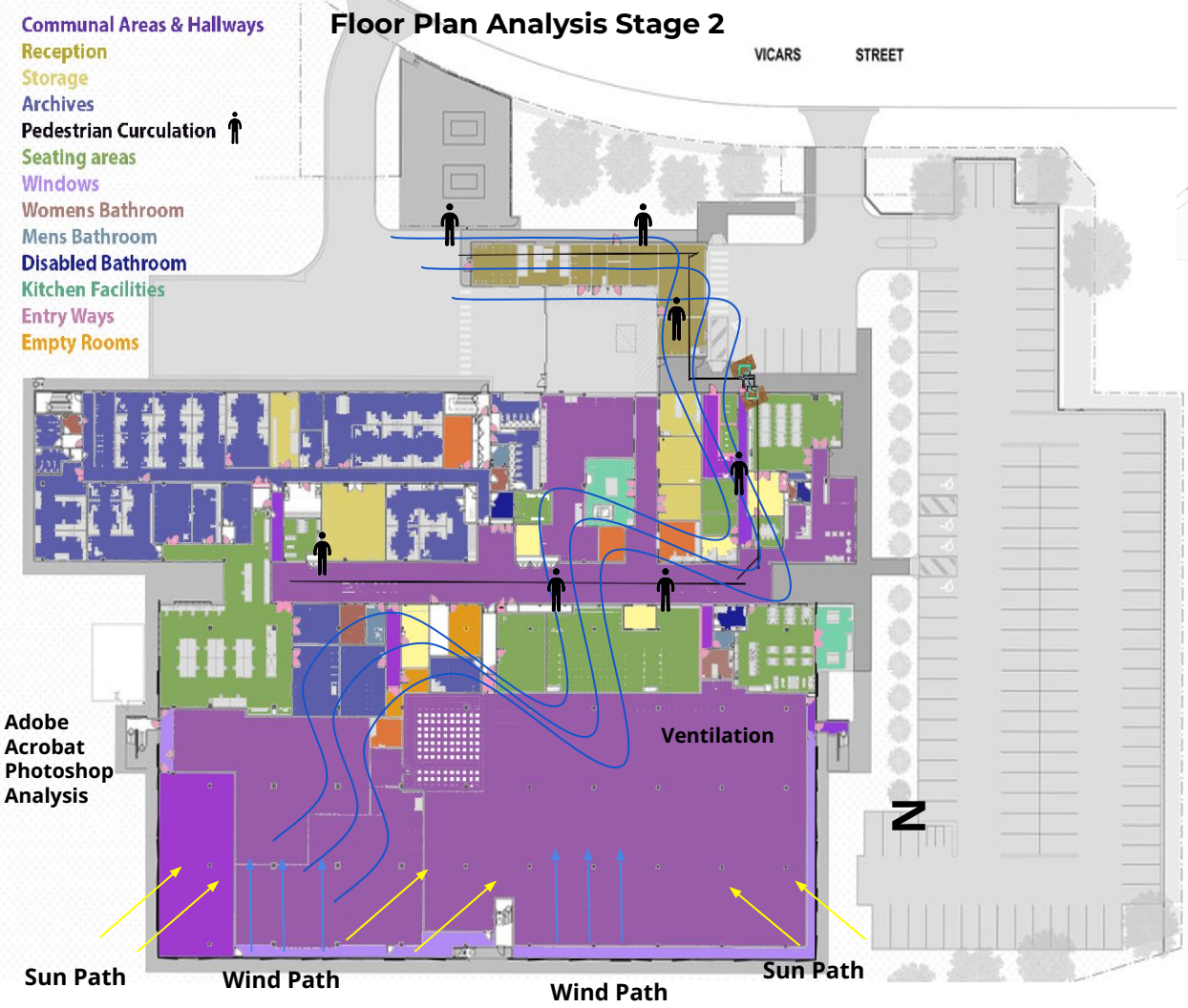
Road Access: The facility is accessible by road including freeways and local roads, with designated entrances and parking areas for visitors and staff. Roads leading to the facility may are well-marked and easily navigable, allowing for convenient access by car or other private vehicles.

Natural Light: The site benefits from abundant natural light throughout the day due to its open exposure and minimal obstruction from surrounding buildings or vegetation. During daylight hours, sunlight penetrates deeply into the site, creating well-lit outdoor spaces and reducing the need for artificial lighting.Although, as it is not obstructed by vegetation during the hottest part of the day, it may require other elements to the building to ensure that it does not get too hot within the building.



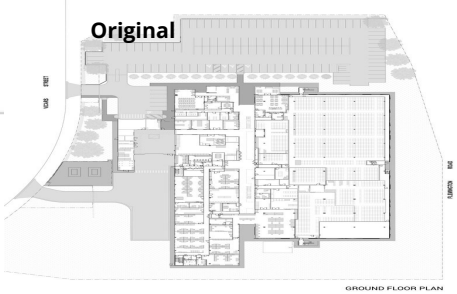
Floor Plan Analysis Stage 2

Communal Areas & Hallways
Reception
Storage
Archives
Pedestrian Curculation
Seating areas
Windows
Womens Bathroom
Mens Bathroom
Disabled Bathroom
Kitchen Facilities
Entry Ways
Empty Rooms



Adobe Acrobat Photoshop Analysis

Original



Spatial Relationships: Administrative areas are conveniently situated near the entrance for easy accessibility by employees and visitors. Conversely, storage and loading zones are located towards the rear or side of the building, minimizing disruption to office operations.

Natural Light and Views: Incorporating large windows or skylights into the design introduces natural light into the warehouse space, reducing reliance on artificial lighting and enhancing the overall workspace ambiance.

Furniture Placement: Office furniture within the administrative areas of this warehouse are thoughtfully arranged to maximize space utilization and functionality. Desks, chairs, and filing cabinets are strategically positioned to facilitate workflow and foster effective communication

Room Layout and Functionality: Within this warehouse, the primary floor area is designated for storage and archive inventory management, featuring expansive, open space with high ceilings tailored for efficiently storing large quantities of goods and materials.

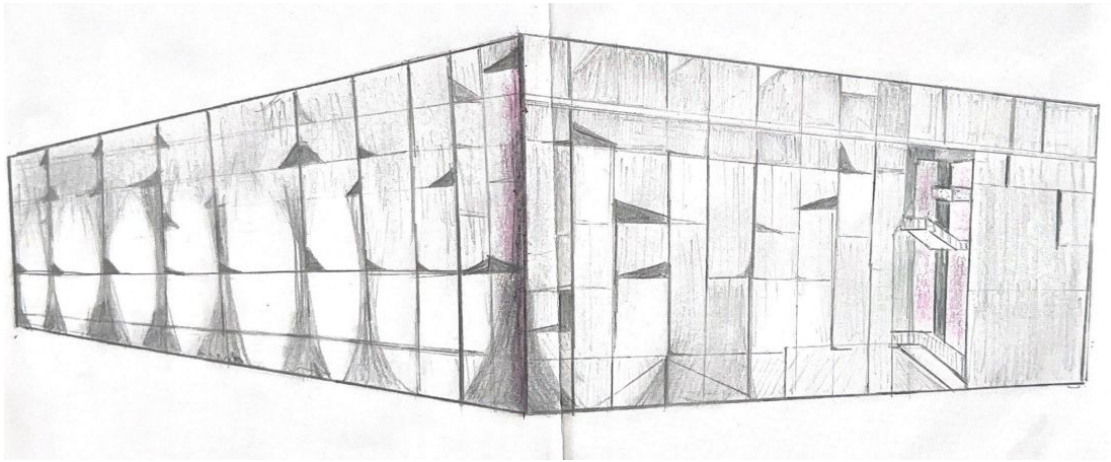
Circulation: The layout of this warehouse prioritizes efficient circulation pathways to facilitate the smooth movement of people and goods throughout the space. Wide aisles and unobstructed pathways are meticulously designed to accommodate the passage of forklifts, pallet jacks, and other equipment essential for warehouse operations. Additionally, strategic placement of loading docks and access points ensures seamless ingress and egress of goods, minimizing congestion and optimizing workflow. By prioritizing circulation in the floor plan design, the warehouse promotes productivity, safety, and operational efficiency within its dynamic environment.

Floor Plan Analysis Continued.

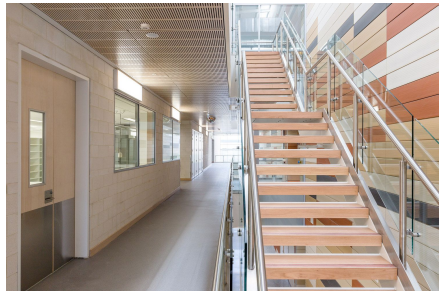
Wind and Sun Path Analysis: The warehouse-style building is designed with sustainability in mind, incorporating an analysis of wind patterns and sun paths. By strategically orienting the building and maximizing natural ventilation and daylight, it minimizes energy consumption and promotes occupant comfort. This approach reduces reliance on artificial lighting and cooling systems, aligning with principles of environmentally conscious architecture.

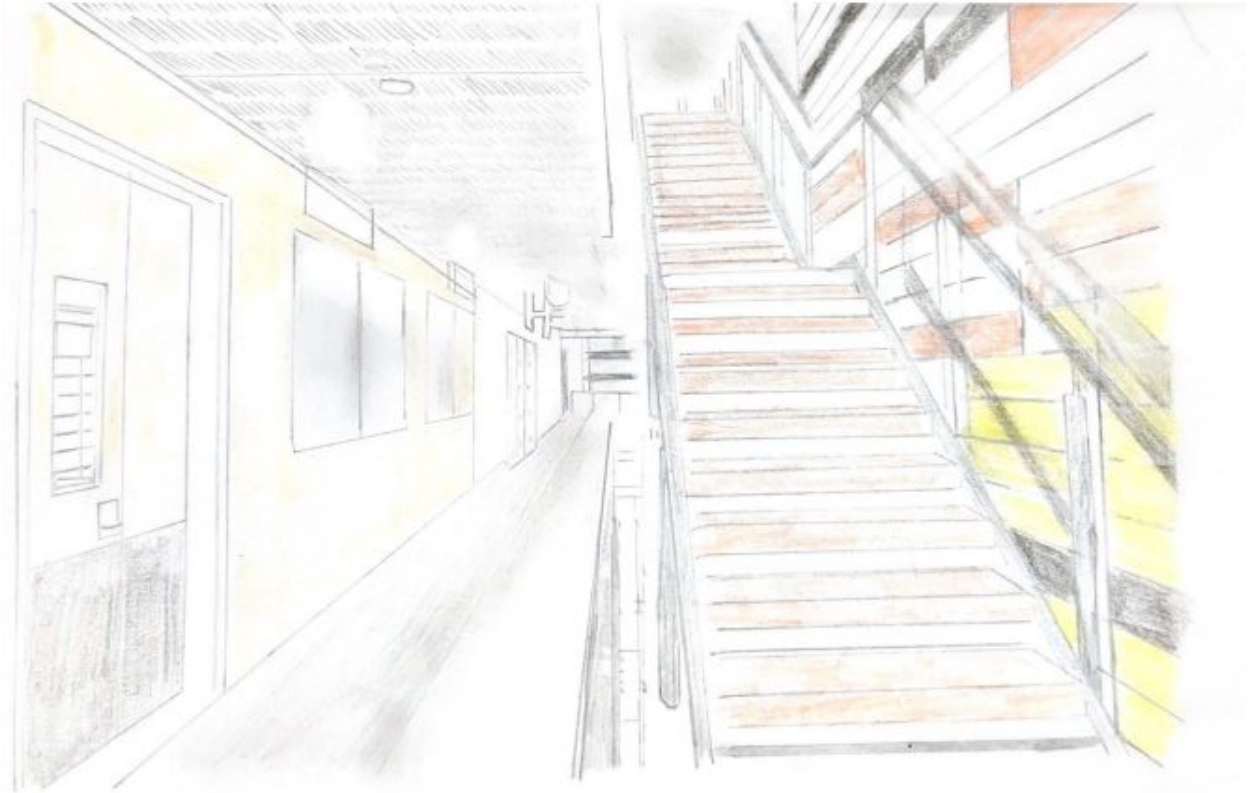
Ventilation Optimization: The warehouse-style building prioritizes ventilation to enhance indoor air quality and occupant comfort while minimizing energy consumption. Through strategic placement of windows, vents, and airflow pathways, the design facilitates natural ventilation, reducing the need for mechanical systems and promoting energy efficiency. Additionally, the layout and orientation of the building is optimised to capture prevailing winds and promote airflow throughout the space, further enhancing ventilation effectiveness. By integrating ventilation optimization into the design process, the warehouse fosters a healthy and sustainable indoor environment conducive to productivity and well-being.

Hand Sketching Stage 1



Hand Sketching Stage 2





Plans and Timeline for my model of the National archives Preservation

Shopping List:

- Foam Board (A1 5mm) $9.54
- CardBoard (3mm 900x600mm) $4.96
- Clear Contact Paper (5mm) $5.85
- Clear Gorilla Glue (51ml) $12.95
- Gorilla Emergency Repair Tape, 8.2 Meter Length, Clear $19.43
- Coloured Markers $2.00
- Quill A4 280gsm Kraft Board 25 Pack $31.29
- Quill A3 Flute Board Assorted 5 Pack $13.49
- Fiskars Softgrip Craft Knife $17.39

Utensils at home:

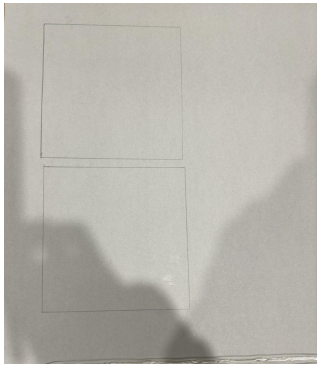
- Scissors
- Ruler
- Stencil
- Stanley Knife
- Sharpener
- Eraser
- Pencils

Plan:

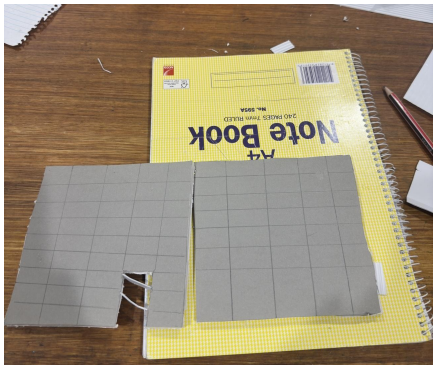
1. Utilizing printed drawings to trace and replicate shapes and structures onto foam board.
2. Employing kraft board as the foundation/base material for the model.
3. Crafting the model by shaping foam board into desired forms.
4. Colour in the cut outs to replicated colour pallet.
5. Adhering primary structural components together using glue.
6. Incorporating clear contact paper sheeting for windows.
7. Utilizing silver flute board to represent metal roofing materials
8. Allocate additional days for the adhesive to fully dry.
9. Completion of Model.

Time Management	Date	Completion Status
Materials	31/03/2024	✅
Started Marking out measurements on Foam Board	31/03/2024	✅
Cutting out parts of model	31/03/2024	✅
Assembling Model	02/04/2024	✅
Final touches and colour selections	04/04/2024	✅
Finish Project	12/04/2024	

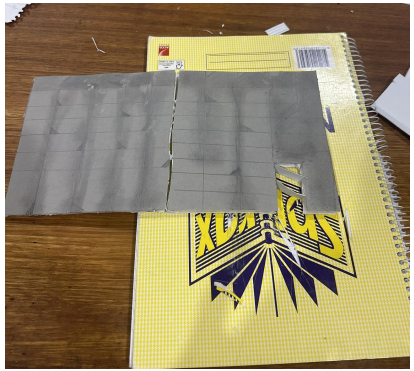
Physical Model Progress



Transferring the outlines and dimensions of wall and concrete panels onto foam board.

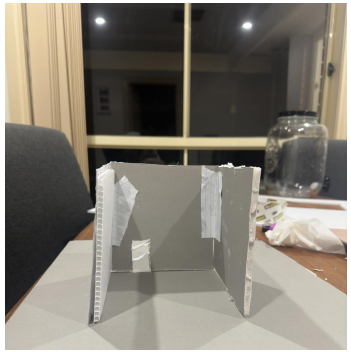


Indicating and marking out the areas where dimensions will be shaded and constructing the side access stairway, making sure they fit in the opening created.

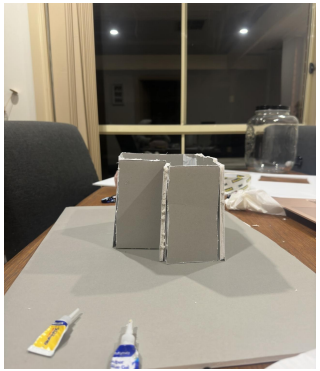


Rendering the facade to replicate the textured appearance depicted in sketches and reference photos.

Physical Model Progress Continued.



Affixing the concrete panels together with super glue and connector paper to guarantee their stability and cohesion.
As depicted above, showcasing the final outcome with securely assembled walls, exemplifying the successful application of super glue and connector paper to ensure structural integrity.

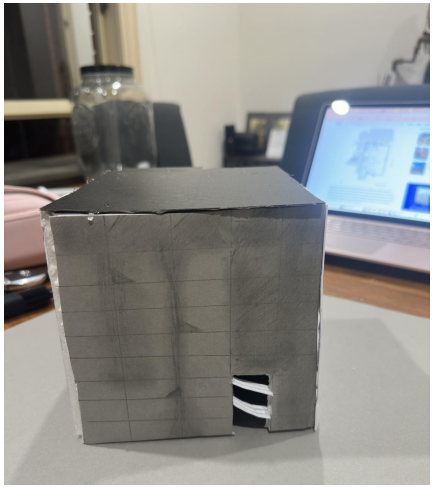




Physical Model Progress Continued.



Constructing the rear window wall using plastic material to serve as the windows, and exhibiting the precise cutouts for the windows as illustrated in the image above.



Evaluating the roof to verify the accuracy of measurements for the roofing covering over the entirety of the building.



Upon the completion of securing the roof, the next step entails allowing it to dry thoroughly. In the ensuing days, I will commence adding final touches and accents to the project.

Reflection and Plan for my Model Stage 1

As I delve into constructing my model, I've come to realize that achieving precision and authenticity demands thorough groundwork and unwavering commitment to each phase of the endeavor. The journey commences with extensive research, delving into architectural blueprints and reference materials to grasp the intricate dimensions and characteristics of the structure, ensuring meticulous adherence to scale. This foundational understanding informs my material selection, ensuring alignment with the intended scale and level of intricacy. Armed with precise tools like rulers and cutting implements, I meticulously measure and shape each component, striving for immaculate lines and exact angles. The assembly process unfolds methodically, commencing with the groundwork and steadily layering components while regularly consulting my research to uphold fidelity. Patience becomes paramount, allowing adequate drying time for adhesives and reinforcing structural integrity as needed. Throughout this journey, I continuously reference my research, making adjustments to maintain faithful representation. With each stroke and detail, I endeavor to encapsulate the essence of the structure, transforming it into a tangible manifestation of my creative vision. Through this hands-on experience and unwavering dedication to authenticity, my model transcends mere craftsmanship—it becomes a testament to my passion for architectural precision and meticulous detail, all carefully rendered to scale.

Short Reflection/ Graphic Style Plan Stage 1

Choosing Global Industrial Architecture was the focus of our project, a decision fueled by a blend of personal interests and academic aspirations within our group (Maisie, Ornela, Sam and Noor).

In our collaborative efforts, we consciously opted for foam board as the primary material for our model, driven by a collective appreciation for its minimalist aesthetics and its ability to deliver a refined finish. Recognizing the importance of clarity and simplicity in architectural representation, we found that foam board offers the ideal canvas to accentuate the core elements of our design without unnecessary embellishments. Its adaptable nature allows us to effortlessly sculpt and mold it to mirror the intricate details and contours of our vision, ensuring an accurate and visually striking portrayal. By selecting foam board, we ensure that our model will exude a sense of coherence and sophistication, effectively capturing the essence of our design with precision and finesse.

Industrial architecture, with its fusion of functionality, innovation, and historical relevance, offers a captivating narrative of human creativity and progress. It transcends mere structures, delving into the narratives of societal needs, technological advancements, and the dynamic evolution of landscapes. This exploration not only enriches our understanding of architectural evolution but also serves as a mirror reflecting global trends and challenges in construction and engineering. Each industrial site, influenced by its unique cultural, environmental, and economic context, offers a rich tapestry of architectural styles, construction methods, and materials to study and appreciate.

Moreover, our interest in industrial architecture extends beyond its aesthetic and historical value. It presents an intriguing canvas for exploring sustainability and adaptive reuse a subject increasingly pertinent in contemporary construction discourse. By examining successful adaptive reuse projects, we aim to glean insights into repurposing existing structures, minimizing waste, and fostering resilient, environmentally conscious built environments.

Individually, our fascination with industrial architecture varies, reflecting diverse interests within the realm of construction. Some of us are drawn to the historical narratives embedded in industrial sites, while others are captivated by the potential of sustainable design principles or the integration of technology in modern industrial facilities. This project provides an avenue for each of us to pursue our unique passions while contributing to a collective endeavor.

Selecting the National Archives Preservation Facility as my individual case study within our group's broader exploration of industrial architecture around the world was a decision rooted in a blend of personal fascination and strategic alignment with our overarching project goals.

On a personal level, the National Archives Preservation Facility captivates me due to its unique intersection of history, architecture, and function. It's not just a repository of documents; it's a custodian of our collective memory, safeguarding the records that tell the story of our nation's past. The facility's architectural design, with its emphasis on preservation and symbolism, speaks volumes about the importance of heritage conservation in a rapidly changing world.

Moreover, choosing the National Archives Preservation Facility aligns seamlessly with our group's larger mission of exploring industrial architecture on a global scale. While it may seem distinct from traditional industrial structures like factories or warehouses, the facility embodies the principles of industrial architecture in its design, construction, and purpose. It showcases how architectural innovation and engineering prowess can converge to create a functional yet aesthetically compelling industrial space dedicated to the preservation of cultural heritage.

Group Topic Theme & Further Reflection Stage 2

Our exploration of industrial architecture across different regions highlighted the diversity of design approaches, construction techniques, and historical contexts that shape built environments worldwide. Each case study, including my analysis of the National Archives Preservation Facility, provided a unique lens through which to examine the evolution of industrial structures and their impact on society.

My journey commenced with a meticulous examination of the site surrounding the National Archives Preservation Facility. Through site analysis, I discerned not only the physical characteristics but also the historical and cultural context that informed the facility's design and location. By scrutinizing factors such as topography, orientation, and surrounding infrastructure, I gained invaluable insights into how the site's features influenced the architectural decisions and functionality of the facility.

Moving further, my floor plan analysis delved into the intricate details of the floor plan, unraveling the spatial layout and functional organization of the facility. With a keen eye for detail, I dissected each room and space, deciphering their intended purposes and interrelationships. The floor plan served as a blueprint for understanding circulation patterns, access points, and the integration of structural elements, guiding me in comprehending how users navigate the space and interact with its features.

Beyond the physical attributes, my analysis ventured into the realm of architectural symbolism and cultural significance. By scrutinizing the floor plan and architectural elements through a lens of cultural context, I uncovered layers of meaning embedded within the design. From the arrangement of spaces to the choice of materials, every aspect of the facility's architecture resonated with historical narratives and cultural values, reflecting a profound connection to the national identity it seeks to preserve.

Moreover, my analysis extended beyond static representations to encompass dynamic considerations such as sustainability and adaptive reuse. By evaluating the facility's environmental impact and potential for future adaptation, I gained a holistic understanding of its long-term viability and relevance in a rapidly changing world.

In essence, my reflection on the analysis of the National Archives Preservation Facility transcended mere observation, evolving into a nuanced exploration of its historical, cultural, and functional dimensions. Through the synthesis of site analysis, floor plan examination, and cultural contextualization, I uncovered the intricate interplay of factors that shape the built environment and its enduring impact on society.

This journey has been incredibly enlightening, allowing me to contextualize my individual case study within the broader framework of global industrial architecture. It has underscored the importance of understanding the historical, cultural, and functional aspects of industrial structures and their enduring impact on the built environment. Through this collective exploration, I've deepened my understanding of architectural heritage and its significance in shaping the world we inhabit.

Group Topic Theme & Further Reflection Stage 2 Continued.

Exploring the National Archives Preservation Facility has truly broadened my understanding of architecture's role in preserving history and embracing modernity. Each aspect of the facility's design reflects a careful consideration of the past, present, and future.

The precast concrete form, reminiscent of Australia's rugged landscape, serves as a tangible connection to our heritage while prioritizing functionality through thermal efficiency. It's a testament to the enduring importance of safeguarding our cultural legacy.

Conversely, the space designated for archivists embodies a forward-thinking approach, utilizing lightweight materials and modular design principles to accommodate the evolving needs of archival work. This juxtaposition between tradition and innovation highlights the facility's adaptability and readiness for future challenges.

Archive Row, the central circulation spine, emerges as more than just a physical connector; it symbolizes collaboration and unity among those tasked with preserving history. It fosters an environment where archivists can come together, share knowledge, and celebrate their vital role in society.

The light-filled atrium, with its clay panels depicting the transition towards digitized records, serves as a visual representation of the evolving nature of archival practices. It prompts reflection on the intersection of tradition and progress, reminding us of the importance of embracing technological advancements while honoring our past.

In essence, the National Archives Preservation Facility embodies the essence of global industrial architecture by seamlessly blending tradition with innovation. It stands as a testament to the enduring significance of preserving our cultural heritage in an ever-changing world.

Group Theme

In line with our group's vision for simplicity and clarity, our presentation format adheres strictly to a black and white minimal aesthetic. Each slide and visual element has been meticulously curated to maintain a clean, uncluttered appearance, reflecting our commitment to effective communication and streamlined design. By embracing this minimalist approach, we aim to focus attention on the essential content of our presentation, ensuring that our message is conveyed with maximum impact and clarity.